DSC602: Data Visualization

What is Data Visualization?

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Outline

- Introduction
- Some notes

Introduction

- What is data visualization?
  - The practice of translating raw information into a visual context – such as maps or graphs – to make data easier for the human brain to understand and pull insights from
  - Goal
    - To enable our brains to identify patterns, trends, and outliers that are often invisible in raw numbers
      - Cognitive Efficiency: Reduces the mental effort required to process large datasets
      - Communication: Acts as a universal language for technical and non-technical audiences
      - Decision Support: Helps stakeholders make faster, smarter, and more informed strategic choices.

Introduction

- Some definitions for data visualization
  - The visual representation and presentation of data to facilitate understanding
  - Visual representation of datasets designed to help people carry out tasks more effectively
  - The use of computer-generated, interactive, visual representations of data to amplify cognition
  - Creation of images that convey salient information about underlying data and Processes
  - Communication of information using graphical representations

Introduction

- A visualization should
  - Save time
  - Have a clear purpose
  - Include only the relevant content
  - Encode data/information appropriately

Some Notes

- Visualization is suitable when there is a need to augment human capabilities rather than replace people with computational decision-making methods
- The design space of possible visualizations of data is huge
- Visualization design is full of trade-offs, and most possibilities in the design space are ineffective for a particular task
  - Hence, need to validating the effectiveness of any design, but this is difficult
- Visualization design should consider resource limitations viz.
  - Limitations of computers, of humans, and of displays

Some Notes

- Why are humans involved
  - Visual representation of datasets designed to help people carry out tasks more effectively
  - Visualization allows people to analyze data when they don’t know exactly what questions they need to ask in advance
- Why external representation
  - Replaces cognition with perception
    - External representations (e.g., carefully designed images) augment human capacity by allowing us to surpass the limitations of our own internal cognition and memory

Some Notes

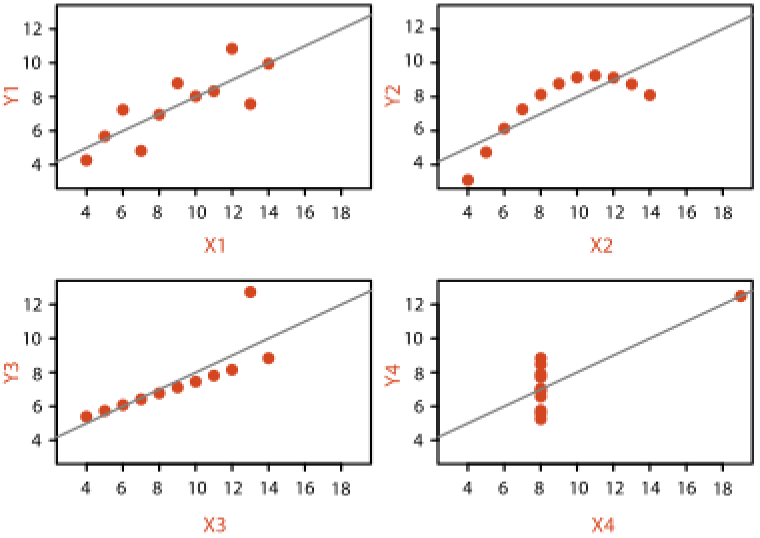
- Why depend on vision
  - Vision is high-bandwidth interface with our brain
  - Overview is possible due to background processing
    - We see everything simultaneously, and the visual system processes in parallel and pre-attentively
  - Sound has lower bandwidth
    - Poorly suited for providing overviews of large information spaces compared with vision

Some Notes

- Why Show the Data in Detail?
  - Visualization tools help people in situations where seeing the dataset structure in detail is better than seeing only a brief summary of it
  - Examples
    - When exploring the data to find patterns, both to confirm expected ones and find unexpected ones
    - When assessing the validity of a statistical model, to judge whether the model in fact fits the data
      - Statistical characterization of datasets is a great, but it has the intrinsic limitation of losing information through summarization (see example on next slide)

Anscombe's Quartet: Raw Data

	1		2		3		4	
	X	Y	X	Y	X	Y	X	Y
	10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
	8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
	13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
	9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
	11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
	14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
	6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
	4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
	12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
	7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
	5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89
Mean	9.0	7.5	9.0	7.5	9.0	7.5	9.0	7.5
Variance	10.0	3.75	10.0	3.75	10.0	3.75	10.0	3.75
Correlation	0.816		0.816		0.816		0.816	



- Anscombe's Quartet
  - Four datasets with identical simple statistical properties: mean, variance, correlation, and linear regression line.
  - However, visual inspection immediately shows how their structures are quite different

Some Notes

- Why Use Interactivity
  - Crucial for building visualization tools that handle complexity
  - Limitations of people and displays precluIde showing all of a large datasets at once
  - Interactive data visualization enables users to directly manipulate graphical representations of data, e.g., by filtering and zooming, to explore complex datasets, uncover hidden patterns, and gain deeper insights in real-time

Methodology

- It can be hard to decide how to approach a visualization problem
- Many computer-based visualization idioms and tools have been created in the past several decades, and considering them one by one leaves you faced with a big collection of different possibilities
- Analysis framework that imposes a structure on this enormous design space and intended to serve as a scaffold to help you think about design choices systematically
- Based on three questions
  - What are we visualising?
  - Why are we visualising it?
  - How can we visualise?

Methodology

- What are we visualising?
  - Major data types and classifications of them
- Why are we visualising it?
  - What is the need for this visualization?
  - Why do the users need this, and what do they need to be able to work with it?
- How can we visualise?
  - The components of a visualization
  - Good and bad practices